	PimpriChinchwad Education Trust's **PimpriChinchwad College of Engineering (PCCoE)** (An Autonomous Institute) Affiliated to SavitribaiPhule Pune University(SPPU)
MDM-Digital Marketing - Capstone Project [BCE27MD06]	

Department: Information Technology **Academic Year: 2026-2027** **Semester: I**

Year: B. Tech **Group ID: [Group ID]** **Date: [DD/MM/2026]**

Problem Statement: Multi-Channel Digital Marketing Attribution and Conversion Analysis (ChannelTrace)

SDG Goals aligned with the project:

SDG No.	Name
SDG 8	Decent Work and Economic Growth
SDG 9	Industry, Innovation and Infrastructure

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Project Guide: Mrs. Minal Shahakar, Dept. of Computer Engineering (Regional Language)

1) Introduction

Background of the Selected Business/Domain

Customers rarely buy a product after a single interaction with a brand. A typical online purchase is preceded by several touchpoints across different digital channels: a person may first see a social media advertisement, later search for the product on Google, open a promotional email, return to the website directly and only then complete the purchase. Businesses of every size now spend across search, social, display, email and referral channels at the same time, yet they usually see only the final sale and not the path that led to it.

Role of Digital Marketing

Digital marketing makes these interactions measurable. Every advertising platform reports impressions, clicks, spend and conversions, and web analytics tools record the channel through which each visit arrives. Marketing attribution uses this data to decide how much credit each channel deserves for a conversion. The answer directly shapes budget decisions: a channel that receives little credit is often cut, even when it plays an important role earlier in the customer journey.

Motivation

Most ad platforms report conversions using their own view, typically last-click, so several platforms can claim the same sale while channels that start journeys appear to contribute little. Advanced attribution tools exist, but they are usually tied to a single platform or require paid subscriptions, and small businesses rarely have the analytical skills to interpret them. The motivation of this project is to study, with real public data, how much a channel's apparent value depends on the attribution rule used, and to turn these methods into a free, easy-to-use tool with an AI assistant that explains the results in plain language.

2) Literature / Market Review

Research Paper 1 – Multichannel Conversion Attribution

Li and Kannan (2014) developed an empirical model of how customers consider and visit a firm's online channels before converting, using individual-level touchpoint data, and validated it with a field experiment. The study shows that the contribution of channels estimated by the model differs considerably from what simple metrics such as last click suggest, because channels influence each other through carryover and spillover effects. This directly supports our research question on whether channel importance changes under different attribution approaches.

Research Paper 2 – Data-Driven Multi-Touch Attribution

Shao and Li (2011) proposed data-driven multi-touch attribution models that learn the contribution of each advertising channel from both converting and non-converting user histories, instead of relying on fixed rules. They also argued that an attribution model should be judged on both accuracy and stability. This work motivates the inclusion of data-driven models (Shapley value) alongside rule-based models in our project, and our use of stability checks such as bootstrap confidence intervals.

Research Paper 3 – Graph-Based (Markov) Attribution

Anderl, Becker, von Wangenheim and Schumann (2016) introduced a graph-based framework that represents customer journeys as Markov chains and measures each channel's importance through its removal effect, that is, how much the probability of conversion falls when the channel is removed. Applied to large customer-journey datasets, the framework showed that common heuristic models misallocate credit. Our project implements this Markov removal-effect model as one of its two data-driven attribution methods.

Research Paper 4 – Predicting Purchase Intention from Session Behaviour

Sakar, Polat, Katircioglu and Kastro (2019) used clickstream and session information from an online store to predict whether a visit will end in a purchase, comparing several machine-learning classifiers. Their work, and the public UCI dataset released with it, shows that session behaviour such as pages viewed, time spent and page value is strongly associated with purchase. We use this dataset to study which visitor and session characteristics are associated with conversion.

Research Paper 5 – Attribution in Display Advertising

Diemert, Meynet, Galland and Lefortier (2017) studied attribution modelling for real-time bidding in display advertising and released the Criteo attribution dataset of 16.5 million impressions. Their work shows that taking attribution into account improves advertising efficiency, and the dataset allows us to study how credit is spread across repeated advertising exposures and how often last-click credit misses earlier impressions.

Existing Market Solutions

Google Analytics 4 offers attribution reports including a data-driven model, Google Ads and Meta Ads Manager each report conversions attributed to their own campaigns, and commercial multi-touch attribution platforms combine data from several sources. These tools are useful, but each platform naturally reports from its own perspective, the multi-source tools are paid, and none of them explain their results in plain language for a small business owner.

3) Gap Identification

Existing Solutions Provide

Existing tools and research provide several useful capabilities:

- Ad platform reports (Google Ads, Meta Ads Manager) provide spend, clicks and conversions for their own campaigns.
- Web analytics tools provide traffic-source information and, in some cases, built-in attribution reports.
- Academic research provides rule-based and data-driven attribution models and conversion-prediction methods.
- Commercial attribution platforms provide cross-channel views for businesses that can pay for them.

Limitations of Existing Practices

Each ad platform measures performance from its own perspective, usually with last-click logic, so the same sale is claimed several times and channels that start customer journeys are undervalued. Cross-channel attribution tools are usually paid or tied to one ecosystem. Research models are rarely available in a form a small business can use, and even when numbers are available, owners often do not know what action to take.

Identified Gap

The gap is therefore not the absence of attribution methods, but the absence of a free, transparent, research-backed tool that compares several attribution models on a business's own data, shows how much the choice of model changes the answer, and explains the results in plain language.

How the Proposed Project Addresses the Gap

The project first studies multi-channel journeys, attribution and conversion behaviour on public datasets. It then packages the same methods into ChannelTrace, a free web application where a business uploads a Google Ads, Meta Ads or customer-journey CSV and receives an insight dashboard, a side-by-side comparison of seven attribution models and an AI assistant that answers questions from the computed results. The proposed analysis flow is:

Upload / Collect Data → Clean → Reconstruct Journeys → Compare Attribution Models → Analyse Conversion → Explain with AI Assistant

4) Problem Statement

Customers interact with several digital marketing channels before purchasing, but businesses usually measure channel performance with single-platform, last-click reports. As a result, budget decisions may favour channels that close sales and undervalue channels that start customer journeys. Existing cross-channel attribution tools are often paid or difficult for small businesses to use and interpret.

The project aims to analyse multi-channel customer journeys using public digital marketing and e-commerce data, compare how different attribution models assign conversion credit, identify behaviours associated with conversion, and deliver these methods as a free plug-and-play dashboard with an AI assistant for businesses.

Target Users: Small and medium businesses running online advertising, digital marketers and marketing students.

5) Objectives

1. To collect, clean and document public datasets on web analytics, advertising and online shopping behaviour.
2. To reconstruct customer journeys and identify common channel paths that lead to a purchase.
3. To compare seven attribution models (first touch, last touch, linear, time decay, position based, Markov chain and Shapley value) and measure how channel credit changes between them.
4. To identify visitor and session behaviours associated with conversion using statistical tests and machine learning.
5. To develop a free plug-and-play dashboard that analyses Google Ads, Meta Ads and customer-journey CSV files.
6. To develop an AI assistant that answers questions and suggests actions using only the computed results.
7. To evaluate the tool for correctness, usability and assistant accuracy, and to document the work as a research paper.

6) Scope of the Project

- **Data Collection and Preparation:** Use the Google Analytics 360 sample, UCI Online Shoppers, Criteo Attribution and Facebook Ad Campaign datasets, each for the question it can support; clean and document every step.
- **Customer Journey Analysis:** Order each visitor's sessions by time and identify the most common paths to purchase.
- **Attribution Modelling:** Implement and compare five rule-based and two data-driven attribution models.
- **Conversion Analysis:** Compare converting and non-converting sessions and model which factors are associated with buying.
- **Insight Dashboard:** Present research findings and analyse uploaded Google Ads, Meta Ads or journey CSV files.
- **AI Assistant:** Answer questions in plain language from computed results only, working offline or with a free AI model.
- **Evaluation:** Check output correctness, run a usability survey (SUS) and measure the assistant's accuracy.

7) Tools and Technologies

- **Python, Pandas and NumPy:** For data cleaning, journey reconstruction and analysis.
- **SciPy and scikit-learn:** For statistical tests, attribution models and machine-learning classifiers.
- **Google BigQuery (free sandbox):** For querying the Google Analytics 360 sample dataset.
- **Streamlit and Plotly:** For the interactive dashboard and charts.
- **Generative AI (Google Gemini free tier) and TF-IDF retrieval:** For the AI assistant, with an offline mode.
- **GitHub and Streamlit Community Cloud:** For version control and free hosting of the application.
- **Power BI / Looker Studio:** For an additional business-intelligence style dashboard (planned).

All tools used are free or open source, so the project has no cost.

8) Proposed Methodology

The project will follow a step-by-step research and development approach:

1. Data Collection → 2. Data Cleaning → 3. Exploratory Analysis → 4. Journey Reconstruction → 5. Attribution Modelling → 6. Conversion Analysis → 7. Tool Development → 8. Evaluation → 9. Research Paper

First, the public datasets will be collected and cleaned, and each will be matched to the research question it can genuinely answer. Customer journeys will then be reconstructed from the Google Analytics 360 sample, and seven attribution models will be applied and compared. The attribution code will be validated on synthetic journeys with a known correct answer, by checking that every model distributes exactly the total number of conversions, and with bootstrap confidence intervals. Conversion behaviour will be studied with Mann-Whitney U and chi-square tests and with logistic regression and random forest models evaluated by ROC-AUC.

The same methods will then be packaged into the ChannelTrace web application and AI assistant. Finally, the tool will be evaluated for correctness on real Facebook advertising data, for usability through a System Usability Scale survey, and for the accuracy of the assistant's answers, and the findings will be written up as a research paper.

9) Result and Discussion

Preliminary results on the UCI Online Shoppers dataset (12,205 sessions after removing duplicates) show an overall conversion rate of 15.6 percent. New visitors convert at 24.9 percent against 14.1 percent for returning visitors, although returning visitors make up 85 percent of sessions. Conversion peaks in November at 25.5 percent. Buyers view a median of 29 product pages against 16 for non-buyers, and all ten behavioural measures differ significantly. A random forest predicts conversion with a ROC-AUC of 0.78 from early-session behaviour (0.93 when the PageValues feature, which partly reflects the outcome, is included).

A working prototype of the dashboard and AI assistant has been developed. On the real Facebook Ad Campaign data it finds that 10 percent of spend went to ad sets with no approved conversions. On synthetic test journeys with known channel roles, the data-driven attribution models correctly identify that last-click reporting under-credits channels that start journeys and over-credits channels that close them.

The expected final result is a comparison of how channel credit changes across attribution models on real multi-channel data, a set of plain-language findings for marketers, and an evaluated, free tool that businesses can use on their own data. All findings will be presented as associations, not proof that a channel caused a sale.

10) Conclusion

The proposed project studies how customers move across digital channels before buying and how strongly the choice of attribution model affects the credit each channel receives. By combining public-data research with a free plug-and-play dashboard and an AI assistant, the project connects academic attribution methods with practical marketing decisions.

The approach is feasible because it relies on publicly available datasets and free tools, and it is measurable through clear evaluation criteria. Overall, the project helps businesses look beyond single-platform, last-click reports and make better-informed decisions about where to spend their marketing budget.

11) References

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Guide Approval

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